Team Name : Team Deccans

Team Leader Name : Mr. Mohammed Shaik Sahil

Problem Statement : PS13: Predictive Fault Analytics for Air Gapped Networks.

Team Members

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Opportunity should be able to explain the following:

- **How different is it from any of the other existing ideas?**
  Most existing network tools rely on reactive threshold monitoring, meaning they only send alerts after a link has already failed or hit 90 percent utilization. Our system is entirely predictive. We analyze the rate of change and precursor signals, like rising jitter variance or BGP hold timer stress, to forecast failures before they impact the mission. Furthermore, instead of just sending a raw technical alert, we even provide solutions based on the notes retrieved from RAG Pipelines and remediations in simple understandable language.

- **How will it be able to solve the problem?**
  The system continuously ingests streaming telemetry from the network nodes. Our machine learning pipeline calculates time to failure, and our local Retrieval Augmented Generation (RAG) pipeline matches that prediction with historical incident data to give operators step by step recovery actions .

- **USP of the proposed solution**
  A 100 percent offline, autonomous AI Copilot that guarantees zero internet dependency. By splitting the hardware architecture, we ensure the intense network simulation and the AI inference engine never compete for resources, eliminating out of memory crashes.

List of features offered by the solution

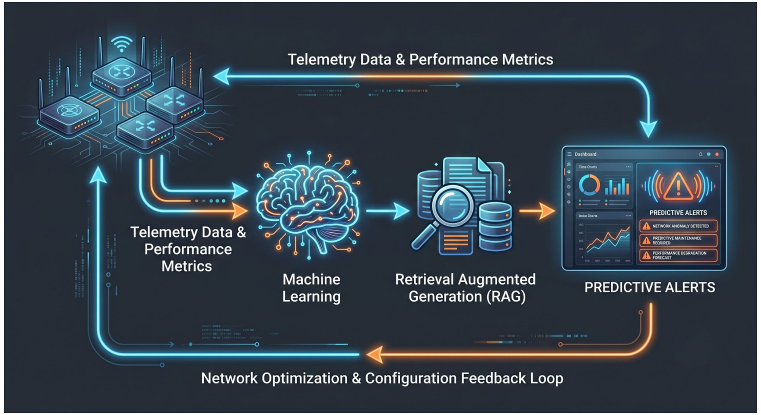
- **Trajectory Based Forecasting:** Predicts exact Service Level Agreement breach times using time series models rather than waiting for static thresholds to be crossed.

- **Graph Based Event Correlation:** Automatically deduplicates alerts by tracing a single root cause to its downstream impacts, significantly reducing alert fatigue.

- **Autonomous Air Gapped Copilot:** A locally hosted, quantized Large Language Model that explains complex network anomalies in plain English without hallucinating fake router names.

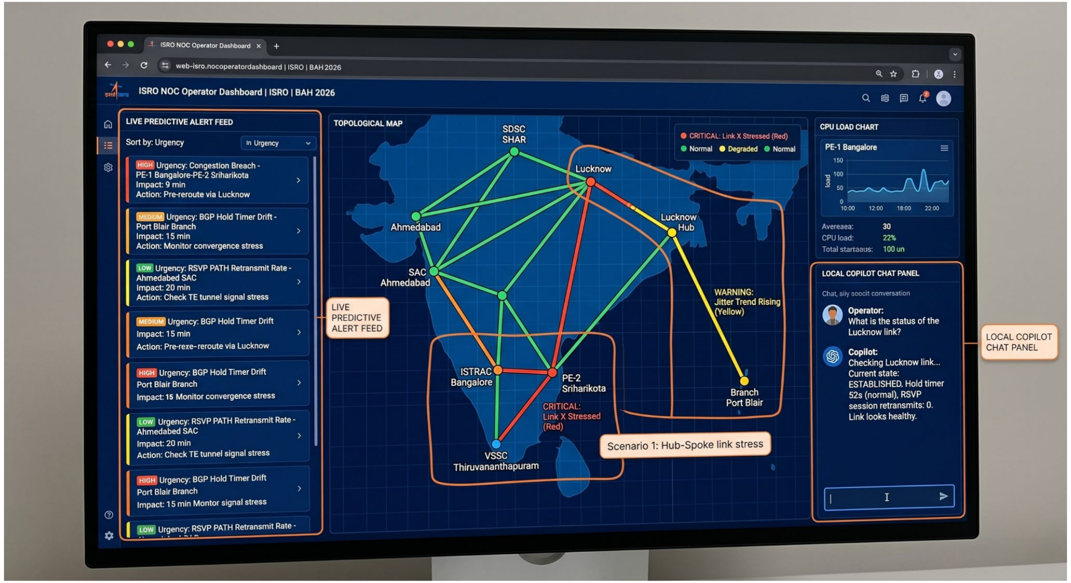
- **Dynamic Alert Prioritization:** Ranks issues dynamically using a compound urgency score based on model confidence and predicted time to impact.

Process flow diagram or Use-case diagram



- **Telemetry Ingestion:** Collect SNMP, syslog, and NetFlow data every 60 seconds.
- **ML Prediction Engine:** Analyze metrics to calculate anomaly scores and predict time to breach.
- **Graph Correlation:** Map the predicted fault to the network topology to identify all affected sites.
- **AI Explanation:** Retrieve relevant offline runbooks and generate a structured root cause explanation.
- **Dashboard Push:** Display the prioritized alert and remediation steps to the network operator.

Wireframes/Mock diagrams of the proposed solution



The operator dashboard will feature a live feed of predictive alerts sorted by urgency, a visual topology map highlighting stressed links in yellow or red, and an interactive chat panel where operators can query the local Copilot for specific link statuses.

Architecture diagram of the proposed solution

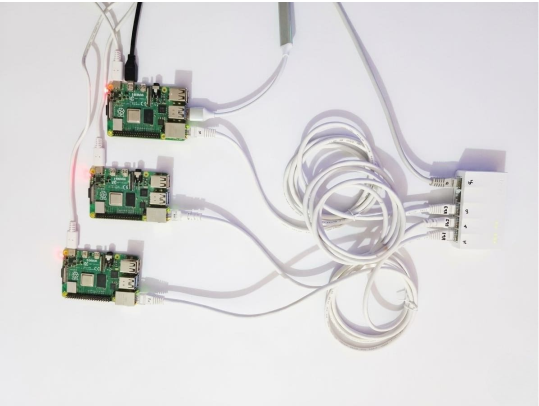


Image 1: Three Raspberry Pi 4 devices acting as Customer Edge and Provider Edge routers, running FRRouting for MPLS label distribution and strongSwan for IPsec tunnels.

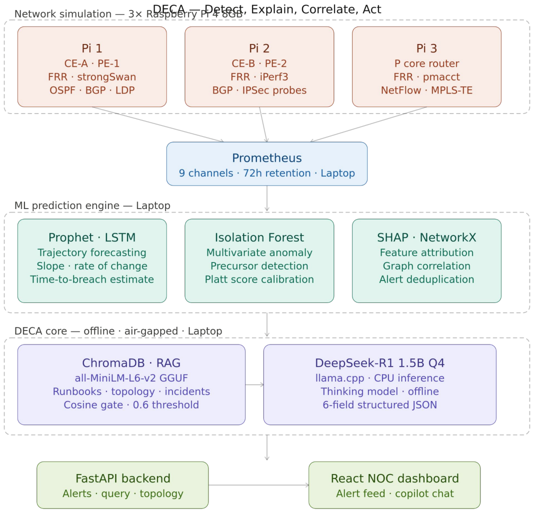


Image 2: The Wireframe for the entire system

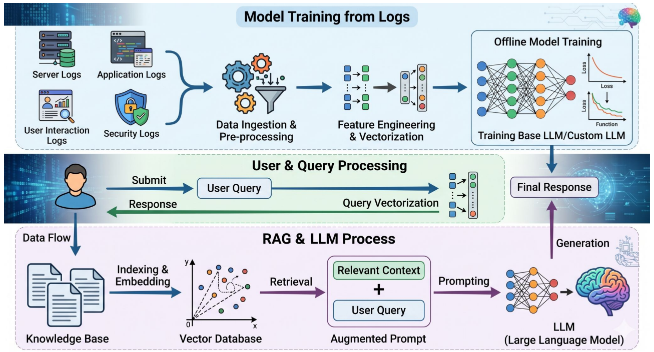


Image 3: The local AI Copilot workflow. It shows the offline RAG process where telemetry logs and runbooks are vectorized, allowing the air gapped LLM to securely answer operator queries with precise, context aware remediation steps.

Technologies to be used in the solution:

- **Network Layer:** Linux network namespaces, FRRouting, strongSwan for SD-WAN simulation.
- **Telemetry & Data:** Telegraf, pmacct, Prometheus Time Series Database.
- **Machine Learning:** Prophet for forecasting, Isolation Forest for anomaly detection, SHAP for feature attribution .
- **AI Copilot:** DeepSeek R1 1.5B quantized model running on llama.cpp, all-MiniLM-L6-v2 for embeddings, ChromaDB for vector storage.
- **Application Layer:** Python, FastAPI backend, React.js frontend.

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THANK YOU